



coreTools Release Notes

coreAssembler - Product Code: 3533-0
coreBuilder - Product Code: 2618-0
coreConsultant - Product Code: N/A

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Contents

- 1 Introduction** **5**
- 2 coreTools W-2024.09-2** **6**
 - 2.1 What's New 6
- 3 Prior Releases** **7**
 - 3.1 W-2024.09-1 7
 - 3.1.1 What's New 7
 - 3.1.2 Issues Resolved 7
 - 3.2 W-2024.09 8
 - 3.2.1 What's New 8
 - 3.2.2 Issues Resolved 8
 - 3.3 V-2024.06-2 8
 - 3.3.1 What's New 8
 - 3.3.2 Issues Resolved 9
 - 3.4 V-2024.06 9
 - 3.4.1 What's New 9
 - 3.5 V-2024.03-1 10
 - 3.5.1 What's New 10
 - 3.5.2 Issues Resolved 10
- 4 System Requirements** **11**
 - 4.1 Supported Platforms and Operating Systems 11
 - 4.2 Hardware Requirements 12
 - 4.3 Software 12
 - 4.4 Licensing 13
- 5 Downloading and Installing this Release** **14**
 - 5.1 Installation Prerequisites 14
 - 5.2 Downloading and Installing the Synopsys Installer 14
 - 5.3 Downloading the coreTools Installation Files 14
 - 5.4 Installing coreTools 15
 - 5.5 Post-Installation Setup 15
 - 5.5.1 Setting up the User Environment 15
 - 5.5.2 Verifying the Tool Installation 15

6	Starting coreTools	16
7	Getting Help	17

Chapter 1

Introduction

This document contains release notes for Synopsys® coreTools release W-2024.09-2. This tool suite consists of the individual design reuse applications coreConsultant, coreAssembler, and coreBuilder.

This document contains information for both system administrators and end-users of the individual tools coreConsultant, coreBuilder, and coreAssembler. You may duplicate this document and distribute it to the intended audience.

The coreConsultant tool guides core integrators through the automated installation, configuration, and synthesis of single reusable synthesizable components.

The coreAssembler tool allows core integrators to assemble multiple components into a subsystem. The tool provides reliable, error-free subsystem netlist generation, design configuration and high-quality synthesis of the subsystem. It also provides auto-connection, register map generation, testbench generation, and simple test builds for DesignWare® Library components.

The coreBuilder tool guides the core developer through packaging a core or design for reuse with coreConsultant or coreAssembler. The coreBuilder tool captures a developer's detailed knowledge of design characteristics and packaging requirements in a technology-independent package, enabling core integrators to easily configure and synthesize the core across a wide range of target technologies.

For detailed descriptions of individual tools, install coreTools as described in the installation procedure in this document, invoke the individual tool, and view the user guide available from the Help menu of each application.

Chapter 2

coreTools W-2024.09-2

This chapter documents changes made in this release. This can include new features, resolved issues, known problems, backward compatability issues, and newly deprecated features. Not all releases will contain updates in all of these areas. Most releases will include additional minor enhancements and bug fixes that are not documented here.

2.1 What's New

This section lists the main changes and enhancements in this release. Releases will typically contain additional minor enhancements that are not documented here.

- General
 - coreTools has been updated to be compatible with the following operating system version
 - CentOS 7.3.1611+
 - RHEL 7.3+; 8+
 - SLES 12.4+; 15-SP2+
 - AlmaLinux: 8.4+
 - Rocky Linux: 8.4+
- STAR ID 5925565*

Chapter 3

Prior Releases

This chapter lists new features and issues resolved in prior releases. Releases will typically contain additional enhancements and bug fixes that are not documented here.

This chapter documents updates for:

- [W-2024.09-1](#)
- [W-2024.09](#)
- [V-2024.06-2](#)
- [V-2024.06](#)
- [V-2024.03-1](#)

3.1 W-2024.09-1

3.1.1 What's New

- IP-XACT
 - Enhanced the IP-XACT reader to accommodate various representations of IP-XACT files with 'ipxact:language=SystemVerilog'
STAR ID 5875048

3.1.2 Issues Resolved

- IP-XACT
 - Corrected an issue in the SCR checker (SCR 7.10) where the SCR checker was incorrectly identifying FieldUseEnumCond failures, resulting in false positives.
STAR ID 5810157
 - Fixed the issue with Design Configuration Generator when the option for parameter passing is enabled, the error led to a malformed node error while writing the design configuration xml.
STAR ID 5840496
- General
 - Resolved tool crash during constraint propagation in coreAssembler.
STAR ID 5847592
 - Introduced a check in Load Designs that displays a warning message when it encounters an unmatched substitution pragma tag in the RTL being analysed.
STAR ID 5858545

3.2 W-2024.09

3.2.1 What's New

- Implementation flow
 - Updated the Tech Cell Bind activity GUI by introducing a check-box to the Tech Cell Bind dialog "Restrict to Cells With Matching I/O Counts".
 - When enabled, the cell selection, drop-down for a given binding, must only list cells with a matching number of inputs and outputs.
STAR ID 4224452
 - Enhanced Technology Cell Binding to support port inversion for both input port and output ports.
STAR ID 5566061
- Synthesis constraints
 - Added -split option to the write_sdc command. When used, the SDC files are partitioned into multiple files based on the constraint type, and each of which are sourced through the top-level SDC file.
STAR ID 5688980
- General
 - Activity RTLStaticChecking does not support components with encrypted source.
STAR ID 5763602
 - auto-generated
STAR ID 5787931

3.2.2 Issues Resolved

- HDL Analysis
 - Resolved error for cast of SystemVerilog value to unsigned.
STAR ID 5701242

3.3 V-2024.06-2

3.3.1 What's New

- HDL Analysis
 - Add support for cast to struct type parameter
 - Sample : parameter struct_t p1 = struct_t('0)
STAR ID 5606760
 - Introduced support for accurately modelling SV RTL (Verilog) code that utilizes anonymous nested structs
STAR ID 5606819
- General
 - Corrected the error in \$bits calculation for typedefs involving multiple packages and using \$clog2
STAR ID 5606821

3.3.2 Issues Resolved

- IP-XACT
 - Updated the IP-XACT writer to resolve a compatibility issue between the 'type' and 'datatype' for module parameters of type 'bit', ensuring seamless integration and accurate data representation.
STAR ID 5580435
- Synthesis constraints
 - When write_sdc is invoked with both -use_clk_expr and -relative, clock constraints defined with absolute integer values are written incorrectly due to use of integer division to calculate percentages.
STAR ID 5650864
- General
 - Resolved error and fix the BitWidth computation for parameter that is set to another bitfield parameter.
Sample :

```
parameter [27:0] bf2 = 28'b1010101;
parameter bf2_ref = bf2;
```

STAR ID 5606757

3.4 V-2024.06

3.4.1 What's New

- IP-XACT
 - Introduced a preference parameter 'interface_to_design_port_width_check' that is valid in core-Builder.
 - When this preference parameter is set to true, the widths of the design and interface ports are checked for consistency before generating the Component IP-XACT file
STAR ID 5524628
- Implementation flow
 - Enhanced the ASIC synthesis workflow by incorporating DFTMAX SEQ CODEC compression capabilities powered by TestMaxManager
STAR ID 5272309
 - Added support for RTL versus RTL comparison in coreAssembler using Formal Verification Activity
STAR ID 5506286
 - Introduced a design attribute to inject custom scripts in the TestMaxManager flow
 - Design attribute 'TmmAuxScript' & 'TmmAuxScriptComment'
 - Insertion points are (pre_read_rtl, post_read_rtl, report, output)
STAR ID 5534414
- General

- Runstyle 'remote' no longer supported.
STAR ID 5501186
- Introduced a Constraint Verification activity that leverages the Timing Constraint Manager (TCM) to verify constraints for the Subsystem, ensuring accuracy and reliability.
 - Consult the coreAssembler User Guide, specifically Section 6.10, for further information.*STAR ID 5557438*

3.5 V-2024.03-1

3.5.1 What's New

- IP-XACT
 - Update the XML design writer to export the hexadecimal value in the `configurableElementValue` element using the 'h' format when writing IP-XACT 1685-2014.
STAR ID 5519748
- Implementation flow
 - Modified the `report_qor.tcl` script in the Fusion Compiler synthesis flow to omit the `report_supply_set` function call when the `EnableUPFFlow` flag is set to false.
STAR ID 5508633
- HDL Analysis
 - Added support for the Verilog built-in function `$size` during RTL modelling.
STAR ID 5495478

3.5.2 Issues Resolved

- IP-XACT
 - Updated IP-XACT component writer to write the bit type parameters as 'bit'.
STAR ID 5359594
- General
 - Fixed issue with SystemVerilog package prefixing after the prefixed component was moved using `move_components`, `ungroup_components` or `ungroup_components`
STAR ID 5425598
 - Resolved the problem where reopening the workspace failed after the activity Techsetup was complete
STAR ID 5427662

Chapter 4

System Requirements

Before you install the coreConsultant, coreBuilder, and coreAssembler tools, make sure that you have write permission for your *coreTools_install_directory* and that your system meets all the requirements listed below.

4.1 Supported Platforms and Operating Systems

Table 1-1 lists the supported platforms, operating systems, and corresponding Synopsys platforms for this release. Many platforms require operating system (OS) patches. For detailed platform support information, see the Release Specific Support page on the Synopsys Qualified System Configuration website at the following address:

<http://www.synopsys.com/qsc>

Select the appropriate foundation for your release (for example, the foundation for this release is W). This QSC Web page provides release-specific information about supported hardware, operating systems, and required OS patches. If the required patch described on this page is not available from the platform vendor, install the most recent patch instead. Synopsys products used by coreTools have been verified against the supported platforms listed in Table 1-1.

Table 1-1 Supported Platforms and Operating Systems

Compute platform	Operating System	Synopsys platform keyword	Banner platform keyword	Desktop windowing environment
x86_64	Red Hat Enterprise Linux (RHEL) 7.3+; 8+	linux64 (64-bit mode) ^b	linux64	GNOME
x86_64	SUSE Linux Enterprise Server 12.4+, 15+	linux64 (64-bit mode) ^b	linux64	KDE
x86_64	CentOS 7.3.1611+	linux64 (64-bit mode) ^b	linux64	GNOME

a. Binary-compatible hardware platform or operating system. Note that binary compatibility is not guaranteed.

b. The 64-bit (x86_64) Linux software is binary compatible with Intel or AMD x86_64 processors. (In prior releases, this platform keyword was "amd64".)

4.2 Hardware Requirements

- 850 MB of available hard disk space (per platform, including common files)
- You need web access and a SolvNetPlus ID for authenticated access to the Synopsys download site:

<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>

4.3 Software

To use coreConsultant or coreAssembler to synthesize a core, you must also have the following minimum level of software installed on your system:

- Fusion Compiler, Version Q-2019.12-SP4 or newer
- RTL Architect, Version R-2020.09 or newer
- Design Compiler (NXT), Version P-2019.03 or newer
- Design Compiler (Expert or Ultra), version K-2015.06 or newer
- (V)HDL Compiler, version 2012.06 or newer

The following optional software gives you access to special features and functionality:

- DFT Compiler, version 2012.06 or newer (for scan insertion synthesis)
- TetraMAX, TestMAX ATPG:
 - When using the TetraMax Reference Methodology flow, version L-2016.03 or newer
- TetraMAX:

- For other ATPG flows, version 2012.06 or newer
- Power Compiler, version 2012.06 or newer (for clock gating)
- PrimeTime, version 2012.06 or newer (for timing model extraction, and budgeting)
- Formality, version 2012.06 or newer (for formal verification)
- Physical Compiler, version 2012.06 or newer (for physical synthesis strategies)
- Synplify, version 2012.09 or newer (for FPGA synthesis strategy)
- For VIP / Vera / Simulator tool versions that are supported in coreAssembler when using DesignWare Library components, see the individual IP Release Notes or installation instructions.

4.4 Licensing

All coreTools applications have separate license features.

- coreConsultant does not require a license.
- coreBuilder requires a "coreBuilder" license.
- coreAssembler can use a DWC-coreAssembler, DesignWare, coreAssembler, or coreBuilder license, as follows:
 - coreAssembler pulls a coreAssembler license if available; otherwise, coreAssembler pulls a DWC-coreAssembler, or DesignWare license, in that order. If neither is available, coreAssembler cannot start.
 - When using coreAssembler with a DWC-coreAssembler or DesignWare license, you cannot instantiate or import any non-Synopsys IP.
 - If you attempt to import non-Synopsys coreKit-packaged IP and a coreAssembler license is not available, coreAssembler will also look for, but not consume, a coreBuilder license to allow these component types to be used.

You can access your Synopsys license key file either through the LM_LICENSE_FILE variable or through the SNPSLMD_LICENSE_FILE variable.

Chapter 5

Downloading and Installing this Release

5.1 Installation Prerequisites

- You use the Synopsys Installer to install the coreTools product. For more on the Synopsys Installer, see "**Downloading and Installing the Synopsys Installer**" below.
- It is recommended that you have system administrator privileges.
- You must have write permission in the installation directory.

5.2 Downloading and Installing the Synopsys Installer

You must use the Synopsys Installer to install the coreTools product.

1. If needed, download and install the latest version of the Synopsys Installer; for more, see:
<https://www.synopsys.com/install/>
2. Add the installer directory to your UNIX path.

5.3 Downloading the coreTools Installation Files

If you or your site administrator has not yet downloaded the release image, do so now.

1. Log in to your Synopsys SolvNetPlus account at:
<https://solvnetplus.synopsys.com/>
2. Click on the "Downloads" menu item from the SolvNetPlus home page.
3. From the "My Product Releases" list, choose the product name (coreAssembler, coreBuilder, or coreConsultant). Note that the installer for any of these products installs all three products at once.
4. Choose the appropriate version from the "<product> (coreTools)" selection list.
5. Select your desired mode of downloading the product installation files (HTTPS or FTP).
6. Download the following files:

```
checksum_info.txt
coretools_INSTALL_README.txt
coretools_vW-2024.09-2_common.spf
coretools_vW-2024.09-2_linux64.spf
coretools_vW-2024.09-2_linuxaarch64.spf
```

Note: The coreTools executables are binary compatible across all supported platforms, so only one installation is needed for multi-platform use.

5.4 Installing coreTools

1. To install the coreTools product, enter one of the following installer commands:

```
% installer # To use a text script
```

-OR-

```
% installer -gui (or setup.sh) # To use a GUI
```

2. Answer the installer prompts. To accept the specified default for a given prompt, press Return. To override a default value, enter the desired value and then press Return.

If you encounter installer problems, see the SolvNetPlus article "Troubleshooting Synopsys Installer for Versions 3.x" at:

<https://solvnetplus.synopsys.com/s/article/Troubleshooting-Synopsys-Installer-for-Versions-3-x-1576091229102>

5.5 Post-Installation Setup

5.5.1 Setting up the User Environment

Refer to the appropriate coreTools product for environment setup information:

- Chapter 2: "The coreAssembler Environment" in the [coreAssembler User Guide](#)
- Chapter 2: "The coreConsultant Environment" in the [coreConsultant User Guide](#)
- Chapter 2: "The coreBuilder Environment" in the [coreBuilder User Guide](#)

5.5.2 Verifying the Tool Installation

To use Synopsys tools successfully, you must:

- Run on a supported hardware platform
- Use a supported operating system
- Apply the required OS patches

For general information about supported platforms and operating system patches, go to

<https://www.synopsys.com/qsc>

Chapter 6

Starting coreTools

1. Add the path for your installed coreConsultant, coreBuilder, and coreAssembler executables to your PATH environment variable definition.

```
% export PATH=<coreTools_install_dir>/bin:$PATH
```

2. Invoke the tool. For example, to start coreAssembler:

```
% coreAssembler
```

3. To view the user guide, select **Help > coreAssembler Tool Help > User Guide** from the toolbar menu.

Chapter 7

Getting Help

If you have a question about using Synopsys products, please consult product documentation that is installed on your network. You can also access documentation for DesignWare products on the Web:

- Datasheets for individual DesignWare IP components, located using "Search for IP":
<https://www.synopsys.com/designware-ip.html>
- License Information – Synopsys Licensing QuickStart Guide:
<https://www.synopsys.com/Support/Licensing>

You can obtain product support in the following ways:

- Go to SolvNetPlus to submit support requests and access documentation and software downloads:
<https://solvnetplus.synopsys.com/>
Support requests should be filed with L1 Product = coreTools and L2 Product = coreConsultant, core-Assembler, or coreBuilder, as appropriate.
- Send an e-mail message to support_center@synopsys.com.
- Global Support Centers – contact information and hours of operation:
<https://www.synopsys.com/support/global-support-centers.html>

